Match Lab — Derived Metrics Reference

Everything the app derives from a video-tagging event file (Masar matches), and exactly how each value is calculated.
Engine: `fap.matchlab` · version 2026.10.16 · pitch 105 × 68 m

1. Coordinate frame & pitch zones
2. Data cleaning at import
3. Pass end points, receivers & carries
4. Ball-loss location
5. Event-level derived columns
6. Pass taxonomy
7. Shots & xG
8. Possessions
9. Phases of play
10. Counter-press
11. Second balls
12. Team summary metrics
13. Team analysis (stats, build-up, progression, final third)
14. Player metrics
15. Heat-map definitions
16. Enriched Excel export columns

Adjustable settings. The thirds lines (default 35 m / 70 m) and the counter-press window (6–8 s, default 8 s) can be changed in Match Lab → Settings; every metric below that mentions "own / middle / final third" or "the counter-press window" follows those settings. Only each tag's **start time** is used — clip end times are an automatic fixed length, not real durations.

1. Coordinate frame & pitch zones

The tagging tool writes metres on a 105 × 68 pitch in image orientation (y grows downward) and keeps each team's raw direction of play. Every event is converted into one **analysis frame per acting team**:

Item	Definition
x	0 → 105; the acting team always attacks left → right (attacked goal at x = 105).
y	0 → 68; y = 0 is the team's **right** touchline (on a y-up plot the right flank is at the bottom).
Attack direction	Detected per team and half — never assumed: (1) mean x of the team's shots (≥ 2 shots); else (2) mean forward displacement of completed passes ending in a receive (≥ 15 passes); else (3) one shot; else (4) opposite of the other team. Stored as `attack_sign`; the export also gives `attack_direction_in_file`.
Thirds	Defensive third x < 35 · Middle third 35 ≤ x < 70 · Final third x ≥ 70 (adjustable).
Channels (5 lanes)	Edges at the penalty-box and six-yard-box widths: Right flank y < 13.84 · Right half-space 13.84–24.84 · Centre 24.84–43.16 · Left half-space 43.16–54.16 · Left flank > 54.16.
Penalty box	x ≥ 88.5 and 13.84 ≤ y ≤ 54.16.
Distance to goal	Straight-line distance to the centre of the attacked goal (105, 34).

2. Data cleaning at import

Step	What it does
Column aliases	Recognises the export's column names (e.g. `pos_x_meters`, `begin_match`, `end`) whatever their exact spelling.
Label normalisation	Labels are split, lower-cased and spelling variants unified (Successfull / Succesful → successful, Unsuccessfull → unsuccessful, Cleareance → clearance, …).
Event type	From the tag name (Pass, Receive, Recovery, Interception, Tackle, Duels, Press, Errors, Ball out, Shot, Saved, Fouls, Dribble, Dribble past, Carry, Block, Clearance, Sub in/out, half start/end).
Outcome	1 = success when a label says successful / won / complete / goal; 0 = failure when unsuccessful / lost / miss; empty otherwise. If both success and failure are tagged on one row, the outcome is resolved from the next event (who has the ball next).
Player number	Shirt number as text; `0` = unknown player (never treated as one person); "0, 20" → 20.
Fouls	A Foul row labelled *Offside* becomes an offside; a Foul labelled *Foul won* (and not *Foul committed*) becomes a **foul won** — so it is not counted as a foul committed.
Late receives	A player's Receive stamped up to 2.5 s *after* his own next action (different clip pre-rolls) is moved in front of that action, so the chain reads receive → action.
Label flags	`lab_set_piece`, `lab_transition`, `lab_build_up`, `lab_regular_play`, `lab_under_pressure`, `lab_counter_press` = the label is present on the row.

3. Pass end points, receivers & carries

Value	How it is calculated
Pass end (`end_x`, `end_y`)	The export has no target coordinates. A pass ends at the next ball event within 7 rows and 12 s (same half): the team-mate's **Receive** (completed pass); the **opponent's** interception / recovery / tackle / duel / block / save / pass (lost pass); or where the ball went **out** / offside. The passer's own late-tagged events and Press / Sub rows are skipped. If an end point is tagged in the file it is used.
`end_source`	Which rule gave the end point: `receive`, `opponent`, `ball_out`, `offside`, `tagged`, or the next team-mate action when no Receive was tagged.
Receiver	The team-mate whose Receive ends the pass.

Implicit carries	Receive at A followed by the **same player's** next on-ball action at B (same team, same half, within 15 s, no opponent event in between) and A→B ≥ 5 m → a derived Carry A→B (`derived = True`). Tagged Carry rows without an end point get their end from the same rule.

4. Ball-loss location

Ball loss = unsuccessful pass, unsuccessful dribble, *Ball lost* (dispossessed), or an Error labelled *Ball lost*.

Value	How it is calculated
`loss_x, loss_y`	Where the ball was **actually lost**: lost pass → its end point (= the opponent's next action, or where it went out); any other loss → the location of the opponent's next action (within 7 rows / 12 s). Falls back to the event's own location.
`loss_at`	"opponent's next action", "ball out of play", "offside", "tagged end point" or "own location (no opponent action found)".
Used by	Ball losses & regains map, Phases of play map, losses heat map, losses by third in Team stats, own-third / final-third loss maps, player dashboard. Hover still shows where the action was played from.

5. Event-level derived columns

Column	How it is calculated
`length`	Distance start → end in metres.
`angle`	Direction of travel in degrees: 0 = straight forward, + = towards the left, − = towards the right.
`direction`	Forward |angle| ≤ 45° · Backward |angle| ≥ 135° · otherwise Left / Right.
`length_class`	Short < 15 m · Medium 15–30 m · Long ≥ 30 m.
`progressive`	Wyscout definition, passes and carries: the action moves the ball closer to the goal by ≥ 30 m if it starts and ends in the own half; ≥ 15 m if it goes from the own half into the opponent half; ≥ 10 m if both points are in the opponent half.
`third` / `end_third`	Third of the start / end point.
`channel` / `end_channel`	Lane of the start / end point (5 lanes).
`f3_entry`	Starts before the final-third line and ends beyond it.
`box_entry`	Ends in the penalty box having started outside it.
`switch`	Switch of play: a pass that travels ≥ 30 m across AND goes side to side — starts outside the central lane on one side and ends outside it on the opposite side.
`cross`, `through_ball`, `cutback`	Pass carrying that label.
`key_pass` / `assist`	Pass labelled Key pass or Assist / Assist.
`line_break`, `line`	Pass labelled Line break; the line broken from its First / Mid / Last label (First line, Midfield line, Last line).
`height`	Ground / Low / High from the label.
`set_piece_kind`, `set_piece`, `open_play`	Corner, free kick, throw-in, goal kick, penalty, kick-off from the labels (or the Set piece label); open play = not a set piece.
`under_pressure`	Label Under pressure.
`derived`	True for rows created by the engine (implicit carries).

6. Pass taxonomy (`pass_type`)

One type per pass, first match wins: Corner → Free kick → Throw-in → Goal kick → Cross → Cutback → Through ball → Switch → Line-breaking → Long ball (≥ 30 m) → Regular.

7. Shots & xG

Column	How it is calculated
`shot_goal`	Shot labelled Goal.
`shot_on`	Goal, labelled Saved / On target, or followed by an opponent Save (and not blocked / off target).
`shot_blocked`, `shot_off`	Labels Blocked / Off target.
`shot_result`	Goal · Saved · Blocked · Off target.
`body_part`	Head / Left foot / Right foot from the labels.
`xg`	FAP internal xG model (location, body part, set-piece type, whether assisted by a cross / through ball / cutback / pass in the previous 3 events). If unavailable: transparent logistic model on distance and goal-mouth angle (headers penalised); penalties = 0.76. Clipped to 0–0.99.

8. Possessions

Value	How it is calculated
Control action	Pass, carry, dribble, shot, recovery; a successful Receive; an Interception / Tackle labelled Won.
`poss_id`	A new possession starts when a team takes control after the other team had it, or after a dead ball (ball out, foul, foul won, offside, half start / end, or a shot).
`poss_team`, `in_possession`	Owner of the possession; whether the row's team is the owner.
Start type	Regain (first action is recovery / interception / tackle), the set-piece kind, or Open play.
End reason	Shot · Half end · the first dead ball (Out of play / foul / offside) · Lost.
Per-possession stats	duration (s), passes / completed, actions, players involved, max x reached, reached final third, reached box, shot, goal, start / end / furthest third.

9. Phases of play

Column	How it is calculated
`phase` (in possession)	On-ball actions of the team in possession, by location: Build-up x < 35 · Progression 35–70 · Final third x ≥ 70.
`def_phase` (out of possession)	Defensive actions (tackle, interception, recovery, block, clearance, duel, foul, press) by location: High press x ≥ 70 · Mid block 35–70 · Low block x < 35.
`transition`	On-ball action within 10 s of the start of a possession that began with a regain.
`counter_press`	Defensive action within the counter-press window after the team lost the previous possession, or labelled Counter press.
Phase table	Per phase: actions, share %, passes, accuracy %, forward %, progressive (completed passes + progressive carries), line breaks, losses & loss % (all loss types, counted in the phase the team was in), average pass length, actions under pressure, key players.

10. Counter-press

Value	How it is calculated
Loss counted	Every open-play loss (our possession ends with the opponent taking control). Not counted when a dead ball (ball out / foul / offside) or an opponent restart comes before the opponent's first action.
Loss moment / place	The opponent's first action of their possession (where and when they won the ball). Exception: when our last action is an explicit Ball lost, or the opponent's first tagged action is > 3 s later (tagging gap), our last action is used.
Pressure evidence	Our Press tag, our Duel / Tackle, or an opponent action labelled Under pressure. If the file has no Press tags at all, our interceptions / fouls / blocks count too.
Pressed	Pressure evidence inside the window (6–8 s) before any regain; or a **late** counter-press: pressure up to 4 s after the window while the opponent has not yet completed a pass.
Regained	Our next possession starts with an open-play regain (recovery / interception / tackle won) inside the window (or inside the late extension after a counter-press). A restart after the ball goes out is never a regain.
Category	Pressed & regained · Pressed, not regained · Regained without press · Not pressed, not regained.
Other fields	seconds to press, seconds to regain, first presser, regained by, press timing (in window / late), press evidence, regain → shot / final third, opponent shot / reached our final third, third of the loss.

11. Second balls

Value	How it is calculated
First contact (contested ball)	A clearance (Clearance, or an Interception / Tackle labelled Clear); an aerial duel (label Aerial / Head); or a high / long ball (label High or ≥ 30 m) whose first touch within 4 s is contested or that nobody touches. A long ball cleanly received by a team-mate or cleanly intercepted is a won first contact — not a second-ball situation.
Second ball won by	The first team to **control** the ball (receive, recovery, non-clearing interception / tackle, pass, carry, dribble, shot, save) within **5 s** of the first contact. Further duels / clearances inside the window keep the ball loose.
No winner	Ball out / foul / offside before anyone controls it, or no control within 5 s.
What next	Within 15 s and before the opponent controls the ball: did the winner reach the final third / shoot?
Outputs	Won / decided and %, by first-contact type, by third (where the second ball fell), per player, won → final third / shot, lost → opponent final third / shot, average seconds.

12. Team summary metrics

Metric	How it is calculated
Possession %	Our passes ÷ (our passes + opponent passes).
Field tilt %	Our passes in the final third ÷ both teams' passes in their final thirds.
Pass accuracy %	Completed ÷ attempted passes.
Progressive passes / carries	Completed progressive passes; progressive carries.
Final-third / box entries	Completed passes (and carries) that enter the final third / the box.
Long balls %	Passes ≥ 30 m ÷ passes.
PPDA	Opponent passes in their own 60 % of the pitch ÷ our defensive actions in our attacking 60 % (tackles, interceptions that are not clearances, fouls, duels). Lower = more intense pressing.
Defensive action height	Mean x of our defensive actions (m from our goal line).
Passes / possession	Mean passes per possession.
Aerial duels won %	Duels won ÷ duels with a result.

13. Team analysis

Team stats (both teams side by side — count, attempts, success %)

Category	Rows
Possession	Possession share, field tilt, possessions, possessions with 3+ / 6+ passes, avg passes & duration per possession, possessions reaching the final third, ending in a shot.
Passing	All / short / medium / long, forward / backward / lateral, ground / low / high, progressive, line-breaking (by line broken), switches, through balls, under pressure, by third, average length.
Attacking	Shots, on target, goals, xG, shots in the box, headed shots, final-third & box entries (passes + carries), crosses, key passes, dribbles, touches in the box.
Duels	All / aerial / ground duels won (won ÷ won + lost), tackles won, dribbles completed, second balls won, dribbled past.
Defending	Interceptions that won the ball (rest = cleared), recoveries, clearances (incl. cleared interceptions), blocks, presses tagged, PPDA, average defensive action height, losses regained in the counter-press window, shots and xG conceded.
Discipline · Set pieces	Fouls committed / won, offsides; ball kept from corners, free kicks, throw-ins, goal kicks.
Ball losses	Total, by third (where the ball was lost), errors leading to a shot.

Build-up

Value	How it is calculated
Build-up possessions	Possessions that start in the own third.
Exit	First own action that takes the ball out of the own third: a pass / carry from x < 35 ending ≥ 35, or a dribble / receive / shot at x ≥ 35. Method = Long pass / Pass / Carry / other; success; exit lane.
Other	Left our third, reached the final third, ended in a shot, losses in the own third (where lost), who starts build-ups, top passing pairs, goalkeeper distribution (keeper from the squad role codes, else the deepest regular player).

Progression & final third

Value	How it is calculated
Final-third entries	Passes / carries crossing the 70 m line — by method (Carry, Cross, Through ball, Long pass, Pass), by lane, by player.
Lines broken	Line-breaking passes in the middle third by the line broken.
Time to final third	For possessions that reach the final third: seconds and passes from the start of the possession to the first action in / into the final third.
Progressive origins	Start points of successful progressive passes and carries.
Final-third lanes	Per lane: actions, passes / completed, key passes, crosses, shots, losses.
Chance creation & shot sources	Key passes / assists by type; each shot's source (set piece, transition / regain, after a cross, through ball, pass, individual).

14. Player metrics

Metric	How it is calculated
Defensive actions by type	Tackles (won), interceptions (won · rest cleared), duels (won / decided), recoveries, presses (successful / tagged), blocks, clearances, fouls.
Progressive actions	Progressive passes (completed) + progressive carries; progressive receptions (completed progressive passes received).
Carries & dribbles	Carries (tagged + derived), carry distance (m), carries into the final third, dribbles attempted / successful.
Passing	Passes, accuracy %, key passes, crosses, line breaks, receptions.
Shooting	Shots, xG.
Ball losses	Unsuccessful passes and dribbles, Ball lost, errors labelled Ball lost.
Second balls won	Second balls this player controlled first (section 11).
Counter-press	His open-play losses, how many were counter-pressed / regained, times he was the first presser, regains after a loss.
Comparison	Each metric vs the team's best player with ≥ 15 actions (ball losses inverted), so a substitute with one pass is never "best".

15. Heat-map definitions

Heat map	Events used
Activity	Passes, receives, carries, dribbles, shots, recoveries, interceptions, tackles, duels (tagged rows only).
Receptions	Receive locations (if the file has none: end points of completed passes).
Passes made	Pass start points.
Pass destinations	End points of completed passes.
Ball regains	Recoveries, interceptions and tackles labelled Won.
Defensive actions	Tackles, interceptions, recoveries, blocks, clearances, duels, fouls.
Pressing	Press actions.
Ball losses	Where the ball was lost (section 4).
Styles	Smooth density (Gaussian-smoothed counts) or positional zones (3 thirds × 5 lanes, % of the total).

16. Enriched Excel export columns

The downloaded workbook keeps the club's 45-column schema (with `outcome` right after `event_type`, taken from the labels only: successful / unsuccessful / conflicting) and team / match / player IDs from the reference tables, then adds:

Column	Meaning
attack_direction_in_file	left_to_right / right_to_left in the raw file.
x_attack, y_attack, end_x_attack, end_y_attack	Coordinates in the team's attacking frame (section 1).
end_source, receiver_number / _player_id / _name	Section 3.
length_m, angle_deg, direction, length_class	Section 5.
progressive, final_third_entry, box_entry, switch_of_play, key_pass, assist, set_piece, under_pressure, transition, counter_press, in_possession	Flags as 1 / 0 (sections 5, 9).
pass_type, set_piece_kind	Sections 5–6.
start_third, start_channel, end_third, end_channel	Zones of start / end.
phase, defensive_phase, possession_id, possession_team	Sections 8–9.
xg, shot_result, shot_goal	Section 7.
minute_decimal	Match minute as a decimal.

cp_loss, cp_pressed, cp_regained, cp_seconds_to_press, cp_seconds_to_regain, cp_first_presser, cp_category, cp_press_timing, cp_press_evidence, cp_window_s	Counter-press result on the action that lost the ball (section 10).
One column per label	1 / 0 for every label that appears in the file.

Extra sheets: Carries, Possessions, Players, Team summary, Counter-press, Data audit (checks against the raw file) and Info.

Generated from the Match Lab engine source (`fap.matchlab` : loader, metrics, phases, counterpress, secondball, team, player, heat, export). 16 October 2026.